VentSense — 4S Li-ion early-warning and per-cell isolation rig

Power

File: Power.kicad_sch

Pack_Path

File: Pack_Path.kicad_sch

Measurement

File: Measurement.kicad_sch

MCU_IO

File: MCU_IO.kicad_sch

Connectors

File: Connectors.kicad_sch

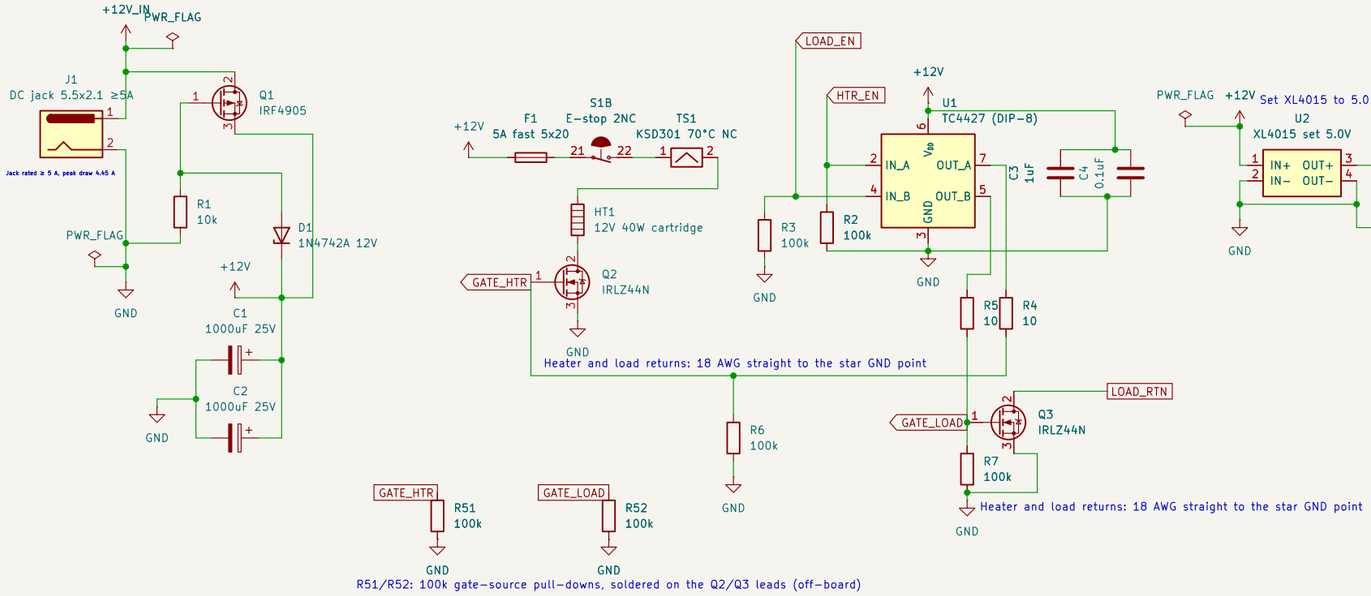
Hierarchy: Power, Pack path, Measurement, MCU and I/O
Bill of materials: bom/VentSense_BOM_Rev2.1.xlsx (Rev 2.1)
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Sheet: /
File: VentSense_Stage2.kicad_sch

Title: VentSense — Stage 2 schematic (root)

Size: A4	Date: 2026-10-20	**Rev: 2.1**
KiCad E.D.A. 10.0.6		Id: 1/6

VentSense — Power



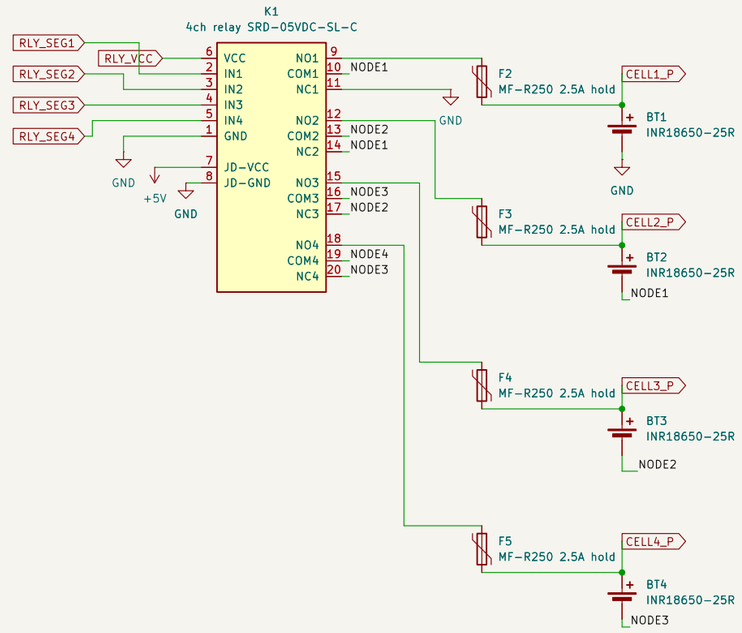
Part of the hierarchical schematic; see the root sheet
Bill of materials: bom/VentSense_BOM_Rev2.1.xlsx (Rev 2.1)
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Sheet: /Power/
File: Power.kicad_sch

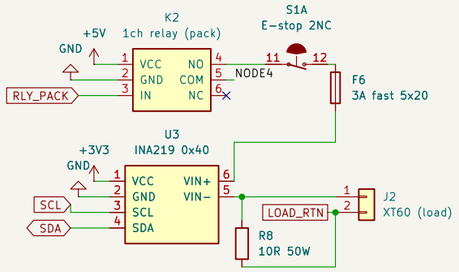
Title: VentSense — Power

Size: A3	Date: 2026-10-20	**Rev: 2.1**
KiCad E.D.A. 10.0.6		Id: 2/6

VentSense — Pack path



Pack negative (cell 1 −) is tied to logic GND at one star point.



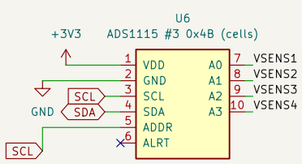
Part of the hierarchical schematic; see the root sheet
Bill of materials: bom/VentSense_BOM_Rev2.1.xlsx (Rev 2.1)
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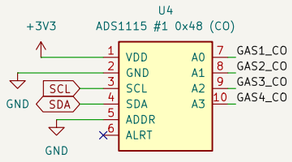
Sheet: /Pack_Path/
File: Pack_Path.kicad_sch

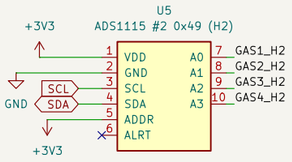
Title: VentSense — Pack path

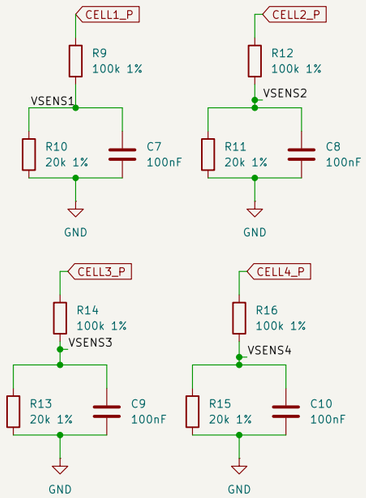
Size: A3	Date: 2026-10-20	**Rev: 2.1**
KiCad E.D.A. 10.0.6		Id: 3/6

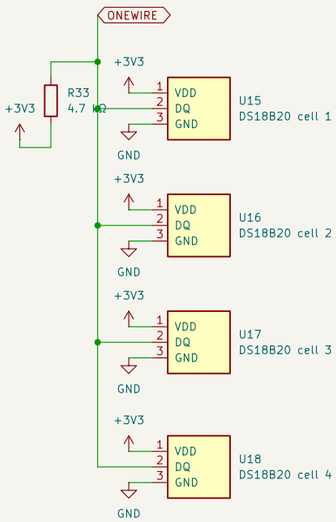
VentSense — Measurement

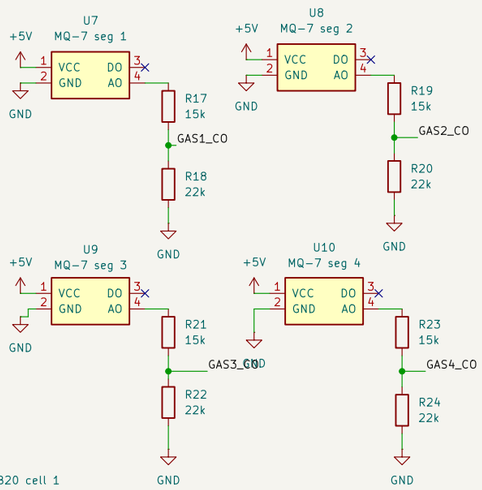


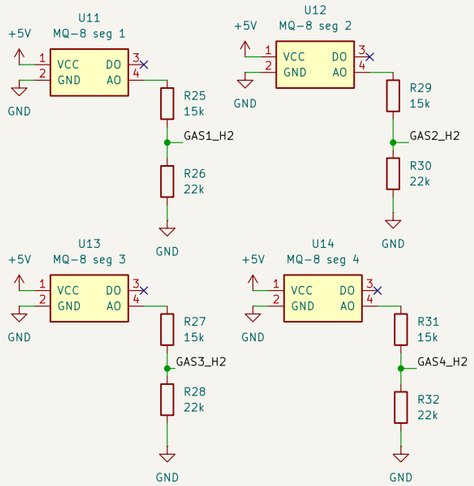












Cell divider inputs carry up to 16.8 V — keep them on their own connector on the carrier PCB.

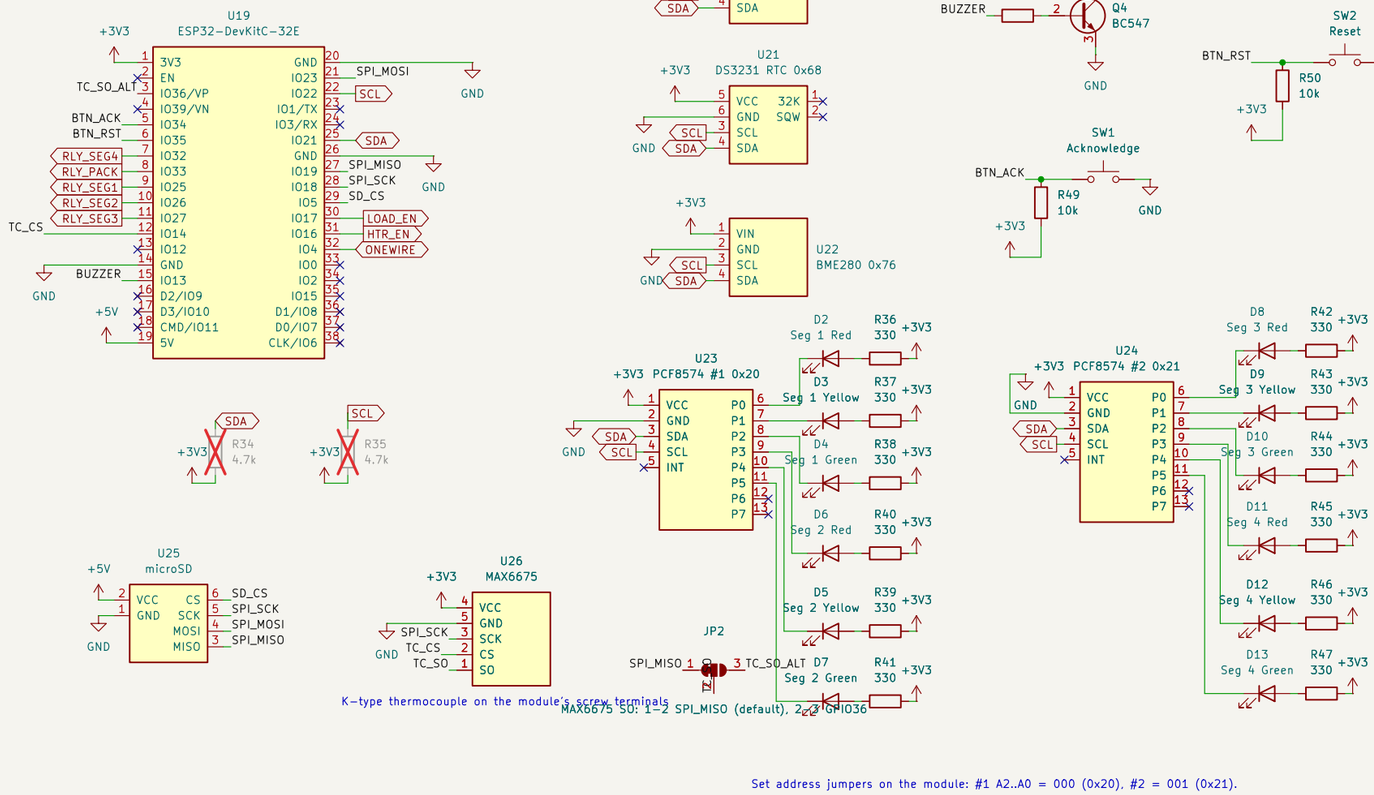
Part of the hierarchical schematic; see the root sheet
Bill of materials: bom/VentSense_BOM_Rev2.1.xlsx (Rev 2.1)
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Sheet: /Measurement/
File: Measurement.kicad_sch

Title: VentSense — Measurement

Size: A3	Date: 2026-10-20	**Rev: 2.1**
KiCad E.D.A. 10.0.6		Id: 4/6

VentSense — MCU and I/O



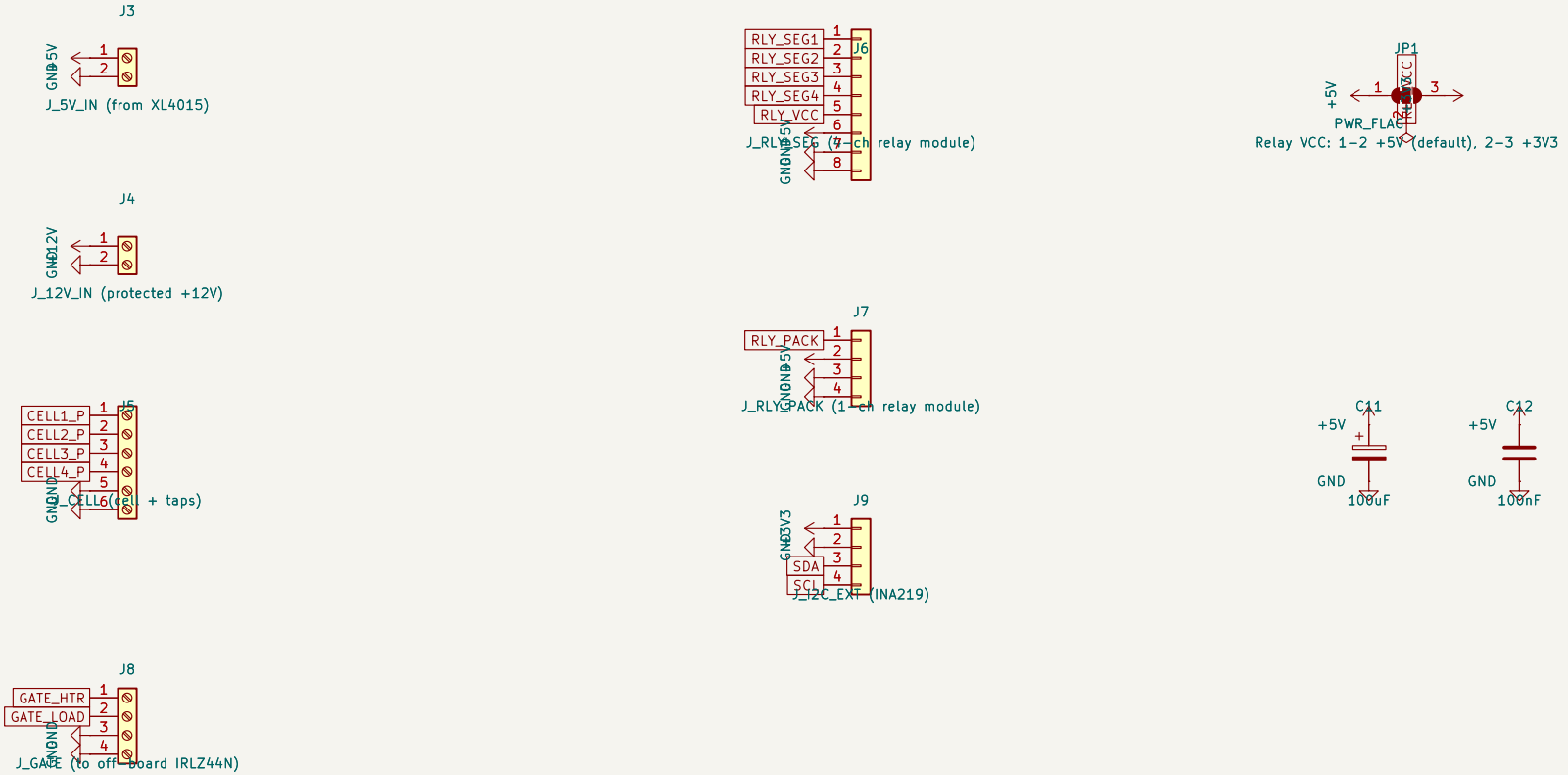
Part of the hierarchical schematic; see the root sheet
Bill of materials: bom/VentSense_BOM_Rev2.1.xlsx (Rev 2.1)
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Sheet: /MCU_IO/
File: MCU_IO.kicad_sch

Title: VentSense — MCU and I/O

Size: A3	Date: 2026-10-20	**Rev: 2.1**
KiCad E.D.A. 10.0.6		Id: 5/6

Off-board wiring: everything that is not on the carrier PCB connects through these connectors. JP1 selects the relay module opto-side VCC (default +5V).



Bill of materials: VentSense_Final_BOM_Gymkhana.xlsx
VK26-IITH-SE-0279
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Sheet: /Connectors/
File: Connectors.kicad_sch

Title: VentSense – Carrier connectors

Size: A4	Date: 2026-11-11	**Rev: 1.1**
KiCad E.D.A. 10.0.6		Id: 6/6